

Capstone Project: Multimodal Semiconductor Automated Defect Classification & Root-Cause Agent (MS-ADC)

This document outlines the scope, architecture, and timeline for an enterprise Capstone Project designed to transition CPSE engineers to the FDE role. Inspired by NVIDIA's modern manufacturing workflows combining **Vision Language Models (VLMs)** and **Vision Foundation Models (VFMs)**, this project demonstrates proficiency across all 32 competencies defined in the FDE Role Profile.

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Company Overview

SiliconPeak Foundries is a simulated leading semiconductor manufacturer producing sub-2 nanometer and advanced packaging chips. As chip density grows, even microscopic wafer defects can ruin entire lots. Historically, SiliconPeak relied on rigid Convolutional Neural Networks (CNNs) for Automated Defect Classification (ADC). However, CNNs require thousands of labeled images per defect class, cannot generalize across new process variations,

and lack semantic reasoning to explain *why* a defect occurred. SiliconPeak wants to modernize fab metrology by deploying an agentic system that combines self-supervised Vision Foundation Models (for die-level optical/e-beam classification) and Reasoning Vision Language Models (for wafer map macro inspection and natural language root-cause diagnosis).

Project Overview

This engagement focuses on establishing a secure, scalable automated defect classification and root-cause analysis agent inside an isolated sandbox. The system bridges edge vision inference with cloud agentic orchestration:

1. **Wafer-Level Reasoning:** Uses Vision Language Models (VLMs like Cosmos Reason / Gemini Multimodal) to interpret macro wafer maps (e.g., center ring, edge ring, scratch patterns), generate natural language root-cause explanations (e.g., chemical contamination vs. mechanical scratch), and support interactive Q&A.
 2. **Die-Level Precision:** Uses Vision Foundation Models (VFM like NV-DINOv2 fine-tuned via NVIDIA TAO Toolkit) for few-shot classification of optical microscopy (OM) and e-beam die images, achieving >98% accuracy with 80% less manual labeling.
 3. **Agentic FMEA RAG:** A central Coordinator Agent links classified defects to internal Failure Mode and Effects Analysis (FMEA) manuals and equipment maintenance logs stored in a vector database, recommending immediate corrective fab actions.
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Assessed Competencies Mapping

To ensure all 32 competencies are demonstrated, this project maps them directly to the following technical deliverables:

1. AI/ML Engineering

- **1. Agentic & Multi-Agent Systems:** Implement a multi-agent orchestration loop (Coordinator Agent, Wafer VLM Specialist, Die VFM Specialist, and Yield RAG Agent) using structured tool-calling and reasoning loops.
- **2. Retrieval & Data Engineering (RAG):** Build a data ingestion pipeline that parses technical FMEA manuals, equipment maintenance PDFs, and cleanroom SOPs into Vertex AI Vector Search with hybrid search and semantic chunking.
- **3. Model Selection & Tuning:** Implement routing logic to send wafer defect maps to reasoning VLMs (Cosmos Reason / Gemini Multimodal) and high-throughput die imagery to optimized VFM endpoints (NV-DINOv2 / TensorRT). Fine-tune VFMs using few-shot learning in TAO Toolkit.

- **4. LLM Ops and Evaluation:** Create an automated evaluation pipeline comparing VLM root-cause explanations against a golden dataset of verified engineer failure logs, tracking faithfulness, BLEU/ROUGE, and classification F1-scores.
- **5. Domain-Applied AI/ML Expertise:** Translate semiconductor yield KPIs (Defect Density \$D_0\$, Die Yield, Escaped Defect Rate) into agent decision thresholds and automated tool definitions.

2. Scoping and Documentation

- **6. Problem Definition:** Author this Capstone Scoping Document defining the transition from high-maintenance CNNs to generalizable VLM/VFM architectures.
- **7. Technical Scope & Constraints:** Document system boundaries (e.g., handling static image datasets for 300mm wafers, out-of-scope physical PLC tool shutdown).
- **8. Stakeholder Alignment & Success Criteria:** Establish clear Definition of Done metrics (>98% die classification accuracy, <3s VLM diagnosis latency).
- **9. System Design Artifacts:** Create architectural block diagrams, sequence flows, and data ingestion pipelines.
- **10. Decision Records (ADRs):** Document trade-offs comparing traditional CNN supervised retraining vs. self-supervised VFM domain adaptation (NV-DINOv2).
- **11. API Documentation:** Deliver complete OpenAPI 3.0 specifications for the FastAPI inspection gateway.
- **12. Operational Documentation:** Write comprehensive runbooks for adding new defect classes via few-shot prompting and handling GPU memory out-of-bounds errors.

3. Security, Privacy, and Compliance

- **13. Authentication & Authorization:** Enforce OAuth2/OIDC service account authentication with Least Privilege IAM roles for cleanroom operators and yield engineers.
- **14. Infrastructure & Network Security:** Isolate inference servers and metrology databases inside a secure Virtual Private Cloud (VPC) with VPC Service Controls.
- **15. Data Protection & Privacy:** Integrate Cloud DLP to ensure proprietary chip layout geometry parameters or fab recipe IP inside prompt metadata are encrypted/redacted.
- **16. AI-Specific Security:** Guard against prompt injection in interactive engineer Q&A and sanitize multimodal image inputs against adversarial noise.
- **17. Compliance & Governance:** Maintain tamper-evident audit logs in Cloud Logging for every automated defect classification and suggested fab recipe tweak.

4. Reliability & Resilience

- **18. Availability Design:** Design multi-replica Cloud Run / Triton Inference Server deployment with health probes and structured fallbacks.
- **19. Observability:** Instrument full-stack OpenTelemetry tracing across the FastAPI gateway, Python agent loop, vector search, and model inference endpoints.

- **20. Failure & Recovery Testing:** Perform chaos testing by simulating inference timeout on the VLM endpoint and verifying automatic graceful degradation.
- **21. Graceful Degradation:** If the reasoning VLM times out during deep root-cause generation, circuit breakers fall back to returning immediate deterministic VFM defect class codes.

5. Performance & Cost Optimization

- **22. Scalability & Elasticity:** Configure horizontal autoscaling for ingestion workers and dynamic batching for die image classification.
- **23. Resource Efficiency:** Optimize edge/cloud compute right-sizing by compiling VFM PyTorch models into NVIDIA TensorRT engines for 4x faster inference at lower GPU footprint.
- **24. AI Cost Management:** Implement prompt and embedding caching for repetitive FMEA manuals and standard reference golden wafers, minimizing token and compute spend.

6. Operational Excellence

- **25. CI/CD & Deployment:** Build automated Google Cloud Build pipelines that run code linting, unit tests, model schema checks, and blue/green container deployment.
- **26. Infrastructure as Code (IaC):** Codify 100% of cloud resources (Cloud Run, BigQuery metrology warehouse, GCS buckets, Vertex AI endpoints, Secret Manager) in Terraform HCL.
- **27. AI Lifecycle Management:** Externalize VLM system prompts and VFM confidence threshold configs into versioned JSON/YAML registries outside application code.
- **28. Testing & Quality Engineering:** Build comprehensive `pytest` suites mocking vision endpoints, verifying mathematical defect density calculators, and testing agent routing logic (>80% coverage).

7. Designing for Change

- **29. Modularity & Abstraction:** Implement strict interface contracts (`DefectClassifierInterface`) allowing seamless swapping between underlying vision providers (e.g., local NIM container vs. cloud Gemini API).
 - **30. Configuration Management:** Use environment separation (Dev/Staging/Sandbox) with runtime feature flags to toggle deep VLM reasoning vs. fast classification.
 - **31. API Design & Versioning:** Contract-first REST API design (`/v1/inspect/wafer`, `/v1/inspect/die`) supporting non-breaking schema evolution.
 - **32. Extensibility:** Create an event-driven webhook callback system that emits standardized defect alerts to downstream fab cleanroom dashboards when critical center-ring defects are detected.
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Project Scope & Deliverables

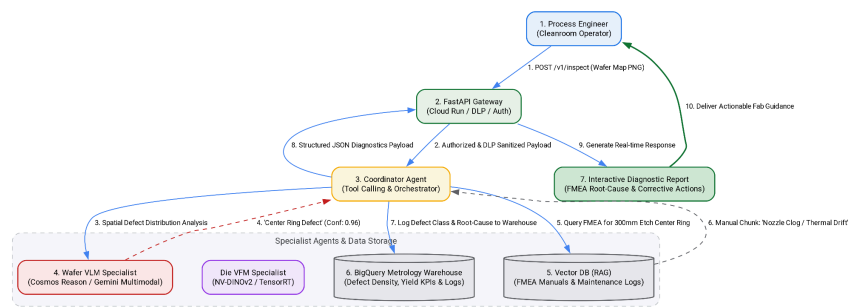
Scope / Deliverable	Responsible Party
Agentic Architecture & Gateway: Multi-agent orchestration system combining VLM reasoning and VFM classification via FastAPI and structured tool-calling.	Google FDE
Vision Foundation Model (VFM) Fine-Tuning Pipeline: Few-shot training workflow using TAO Toolkit and NV-DINOv2 representations for die-level e-beam/optical images.	Google FDE
Infrastructure as Code (IaC): Complete Terraform provisioning of secure VPC, Cloud Run containers, BigQuery metrology dataset, GCS, and IAM roles.	Google FDE
RAG FMEA Knowledge Engine: Vector database pipeline indexing semiconductor failure catalogs, tool maintenance manuals, and recipe tolerances.	Google FDE
Observability & Evaluation Suite: OpenTelemetry tracing, Cloud Logging audit trails, and automated LLM-as-a-Judge yield accuracy benchmarks.	Google FDE

Out of Scope

- Direct physical hardware control, wafer robotic arm manipulation, or automated chemical stop-valve shutdown in live fab equipment.
 - Real-time sub-millisecond edge PLC (Programmable Logic Controller) control loops.
 - Production deployment into SiliconPeak's physical cleanroom manufacturing network (all work performed in Argolis sandbox).
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User Journey

User Journey Flow Diagram

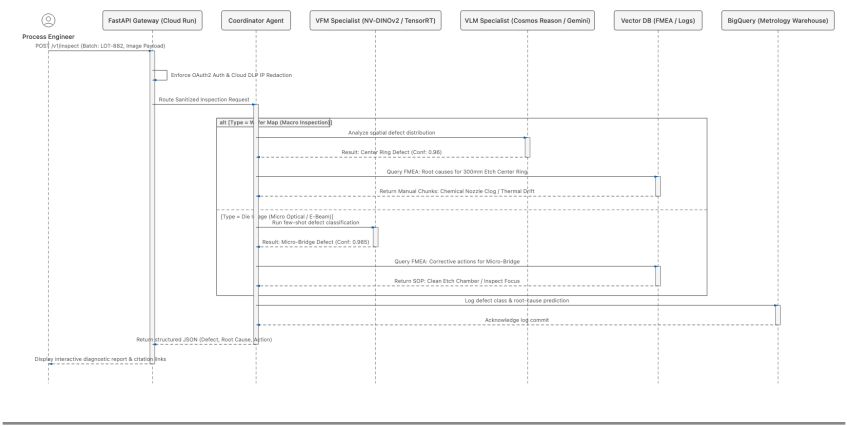


(Visual architecture & sequence flow diagram inserted below)

Sequence Flow Specification

Step	Initiator	Target Component	Action & Payload Description
1	Process Engineer	FastAPI Gateway	POST /v1/inspect with {batch_id: "LOT-882", image: wafer_map.png, type: "wafer_map"}
2	FastAPI Gateway	FastAPI Gateway	Enforces OAuth2/OIDC Auth and Cloud DLP IP Protection on query parameters
3	FastAPI Gateway	Coordinator Agent	Initiates multi-agent inspection workflow with sanitized payload
4	Coordinator Agent	Wafer VLM Specialist	Analyzes spatial defect distribution across macro wafer map
5	Wafer VLM Specialist	Coordinator Agent	Returns defect classification: "Center Ring Defect (Confidence 0.96)"
6	Coordinator Agent	Vector DB (RAG)	Queries FMEA: "Root causes for Center Ring defect in 300mm etch"
7	Vector DB (RAG)	Coordinator Agent	Returns manual chunks: "Chemical spray nozzle clog or wafer chuck thermal drift"
8	Coordinator Agent	BigQuery Metrology	Logs classification, confidence, and root-cause prediction to Metrology Warehouse
9	BigQuery Metrology	Coordinator Agent	Acknowledges database log commit
10	Coordinator Agent	FastAPI Gateway	Returns structured JSON (Defect: Center Ring, Root Cause: Chemical Nozzle Clog, Action: Inspect Chuck/Nozzle)

Step	Initiator	Target Component	Action & Payload Description
11	FastAPI Gateway	Process Engineer	Renders interactive diagnostic report with cited FMEA chunks and historical yield analytics



Technical Requirements

- **Runtime & API Framework:** Python 3.11+, FastAPI, Pydantic v2
- **Vision & AI Models:**
 - **Wafer Macro Reasoning:** NVIDIA Cosmos Reason / Gemini Multimodal (via structured tool calling).
 - **Die Micro Classification:** NVIDIA NV-DINOv2 / TAO Toolkit fine-tuned classifier compiled with TensorRT.
- **Databases & Storage:**
 - **BigQuery:** Structured metrology logs, lot histories, and yield analytics tables.
 - **Vertex AI Vector Search / Cloud SQL pgvector:** Embedded technical manuals, FMEA guidelines, and maintenance logs.
 - **Cloud Storage (GCS):** Raw wafer maps, optical microscopy die PNGs, and golden reference image corpus.
- **GCP & Cloud Infrastructure (Provisioned via Terraform):**
 - **Cloud Run:** Serverless container hosting for the API gateway and multi-agent backend.
 - **Cloud Build & Artifact Registry:** Automated CI/CD build pipelines and container image storage.

- **Secret Manager & Cloud KMS:** Secure storage of API keys, model weights encryption, and OAuth tokens.
 - **Cloud DLP:** Sanitization of proprietary chip design parameters in queries.
 - **OpenTelemetry + Cloud Trace / Logging:** Distributed observability and tamper-evident compliance trails.
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Objectives

Business Objectives

- **Slash Manual Labeling Costs:** Reduce cleanroom engineer hours spent manually categorizing ambiguous defects by 80% through VFM few-shot adaptation.
- **Accelerate Root-Cause Resolution:** Cut average time-to-diagnosis for lot excursions from 6 hours to under 30 seconds by automatically generating FMEA-backed explanations.
- **Eliminate Escaped Defects:** Achieve >98% accuracy on die-level classification, preventing defective dies from entering advanced packaging lines.

Technical Objectives

- **Zero-Hallucination Tool-Calling:** Ensure VLM reasoning outputs are constrained by Pydantic JSON schemas and grounded strictly in verified metrology databases.
 - **High-Throughput / Low-Latency Inference:** Optimize Vision Foundation Model execution using TensorRT to process die inspection batches in under 500ms per image.
 - **Complete Codification:** Guarantee 100% reproducible sandbox provisioning via automated Terraform HCL scripts and Cloud Build CI/CD triggers.
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Deliverable Artefacts

1: Discovery, Scoping & Baseline IaC

- **Objectives:** Finalize project scoping agreement, establish system architecture, and deploy base infrastructure via Terraform.
- **Deliverables:**
 - Approved Scoping Document ([scoping.md](#)) & Architecture Diagrams [Comp 6, 7, 9]
 - Terraform HCL modules provisioning secure VPC, GCS buckets, BigQuery metrology dataset, and IAM service accounts [Comp 13, 14, 26]

- Git repository initialization and development environment structure [Comp 25]

2: Data Ingestion & VFM Few-Shot Setup

- **Objectives:** Ingest wafer/die datasets and configure the Vision Foundation Model classification pipeline.
- **Deliverables:**
 - Python ingestion scripts to load synthetic wafer maps and die images into GCS and BigQuery [Comp 2, 5]
 - ADR documenting the choice of NV-DINOv2 self-supervised feature extraction over scratch-trained CNNs [Comp 10]
 - Fine-tuning script demonstrating few-shot classification setup in TAO Toolkit / PyTorch [Comp 3]

3: Agent Orchestration & RAG FMEA Pipeline

- **Objectives:** Build the multi-agent orchestration loop and index semiconductor failure catalogs.
- **Deliverables:**
 - FastAPI application skeleton with complete OpenAPI 3.0 specs and modular interfaces [Comp 11, 29, 31]
 - Coordinator Agent implementing structured tool calling to route between Wafer VLM and Die VFM tools [Comp 1]
 - Ingestion and embedding pipeline for FMEA manuals into Vertex AI Vector Search [Comp 2]
 - Externalized prompt management configuration files [Comp 27]

4: Security, Guardrails & Automated CI/CD

- **Objectives:** Implement enterprise security controls, PII/IP redaction, and CI/CD pipelines.
- **Deliverables:**
 - Cloud DLP middleware to redact sensitive chip geometry metadata from prompts [Comp 15, 16]
 - Cloud Logging audit logger capturing all automated classifications and operator interactions [Comp 17]
 - `cloudbuild.yaml` running automated formatting, pytest suites, and container builds on pull requests [Comp 25, 28]

5: Reliability, Observability & Inference Optimization

- **Objectives:** Optimize model inference latency, configure observability tracing, and ensure failover resilience.
- **Deliverables:**

- OpenTelemetry tracing middleware exporting request spans and token latencies to Cloud Trace [Comp 19]
- Implementation of prompt/embedding caching for FMEA manuals and golden wafer maps [Comp 24]
- Circuit breakers and fallback logic defaulting to fast VFM classification if deep VLM reasoning times out [Comp 18, 20, 21]
- Load testing scripts confirming Cloud Run autoscaling under simulated cleanroom batch inspection spikes [Comp 22, 23]

6: Evaluation, PRR & Disengagement

- **Objectives:** Benchmark model accuracy, run Product Readiness Review (PRR), and transition code.
- **Deliverables:**
 - Automated evaluation pipeline comparing agent outputs against golden engineering logs (>98% die accuracy, >95% VLM reasoning score) [Comp 4, 28]
 - Product Readiness Review (PRR) readout deck and operational troubleshooting runbook [Comp 8, 12, 30]
 - Webhook callback extension demonstrating real-time fab dashboard alerts [Comp 32]

Verifiable Outcomes & Success Metrics

Success Criteria (Definition of Done)

1. **Functional Verification:**
 - Die-level defect classification accuracy must achieve **>=98.0%** on the evaluation test dataset using few-shot VFM representations.
 - Wafer-level VLM root-cause diagnoses must output 100% schema-valid JSON citing specific FMEA document chunks and matching BQ metrology IDs.
 - Application backend automated test suite (**pytest**) must maintain **>=80%** code coverage.
2. **Operational Verification:**
 - 100% of cloud resources provisioned via automated **terraform apply** with zero manual console configuration.
 - Successful automated zero-downtime deployment to Cloud Run triggered by Git merges to the main branch.
3. **Observability & Latency:**
 - Average end-to-end inspection latency under 500ms for die image classification and under 3.0s for full VLM reasoning + RAG queries.
 - All traces, custom metrics, and audit logs verifiable in Google Cloud Console.

Public Links

Nvidia Blogs and Code

<https://github.com/nvidia-cosmos/cosmos-cookbook>

<https://developer.nvidia.com/blog/optimizing-semiconductor-defect-classification-with-generative-ai-and-vision-foundation-models/>

<http://mirlab.org/dataSet/public/>

<https://www.kaggle.com/datasets/qingyi/wm811k-wafer-map>

This is the exact dataset used in the NVIDIA Metropolis/Cosmos vision pipelines . It provides real-world spatial wafer-bin maps (WBMs) sourced directly from a commercial foundry . The nine failure classes (Center, Donut, Edge-Loc, Edge-Ring, Loc, Random, Scratch, Near-full, and none) exactly match the diagnostic capabilities required by your Wafer VLM Specialist

they may need to data augmentation like we did in micron project for minority defect classes

<https://github.com/Ironbrotherstyle/PCB-DATASET>

<https://www.kaggle.com/datasets/akhatova/pcb-defects>

<https://universe.roboflow.com/nitw-tfufy/pcb-dataset-defect-fa2h3>

While your project targets silicon-die optical microscopy (OM), obtaining public, unclassified micro-die optical images is nearly impossible due to proprietary corporate IP. This dataset serves as an exceptional functional proxy. It contains high-resolution optical images of micro-electronic trace lines